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Successful Execution of Eco-Friendly Cement Solutions for Sustainable Well Integrity

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Abstract

The successful execution of cement operations for low-Portland cement systems is presented. The objective was to lower emissions through reduced-Portland content in cement systems with assurance of full cement integrity in the annulus. The subject well was a heavy casing design gas producer. The planned cement systems applied to a 9 5/8-in. casing inside the reservoir and a 7-in. production liner, both with more than 50% of Portland reduction.

The 9 5/8-in. casing was set inside the reservoir at 8,400 MD and 8,200 TVD. The proposed 9 5/8-in. cement system used a 13.2-lbm/gal reduced-Portland lead slurry and a 15.8-lbm/gal reduced-Portland tail slurry. The 7-in. liner system applied a 15.8-lbm/gal reduced-Portland and resin-based system. The reduction in Portland cement content helped lower CO₂ emissions and the overall carbon footprint. Special spacers placed ahead of the cement helped ensure proper well cleaning, mud removal, and improved cement bond with the formation. Complete laboratory tests for all three systems and all cement parameters were verified.

The 9 5/8-in. casing cement operation was executed successfully, with 75 bbl of lead cement returned to surface and all cement quality parameters confirmed. Cement bond logs (CBLs) showed good-to-moderate cement in the openhole section and 800 ft of cement coverage inside the previous casing.

The 7-in. liner cement operation was also executed successfully, with 20 bbl of cement returned to surface. However, some losses occurred during the cement operation. CBLs indicated good cement in the openhole section.

Cement operation best practices, such as liner rotation, proper centralization, and effective hole cleaning, improved cement operation performance in both sections.

This operation marked the first application of the reduced-Portland system in the country. The project achieved positive results and realized a significant reduction in carbon emissions.

Introduction

In line with the operator's 2030 Sustainability Strategy and the UAE's national decarbonization objectives, this cement operation used a high-performance, low-carbon emission system with less than 50% Portland

cement by mass. This approach directly supported the operator's commitment to reduced emissions and a lower carbon footprint during well construction with preservation of zonal isolation integrity and operational efficiency.

The cement system selected for this well demonstrated superior mechanical properties, such as higher compressive strength at lower densities, improved ductility, and reduced permeability. These characteristics help ensure long-term well integrity and corrosion resistance, particularly in complex formations where equivalent circulating density (ECD) should be carefully managed. Operational deployment of the reduced-Portland slurry system requires no specialized equipment, which allows seamless integration into current rig workflows. This initiative aligns with the operator's "Energy for Life" strategy and confirms its role in using innovative, low-carbon technologies in the upstream value chain.

The project serves as a reference for other operators who seek to balance operational efficiency with environmental stewardship. Adoption of a low-carbon cement system in this horizontal gas well confirmed the operator's leadership in sustainable oilfield practices. The project achieved its objectives of reduced emissions, preserved annular integrity, and optimized operational workflows. This achievement represents a significant step in the operator's journey toward a more sustainable and resilient energy future.

Well Details and Objectives

The candidate well chosen for this initiative is a new horizontal gas producer with a heavy casing design as per shown in Figure 1. The heavy casing design included four casing strings—18 5/8-in. surface casing, 13 3/8-in. casing, and 9 5/8-in. casing—and a 7-in. liner. The sections of interest for this application were the 9 5/8-in. casing and the 7-in. liner. The well was expected to contain CO₂, H₂S, and gas in both sections. These conditions required specialized corrosion-resistant cement systems with advanced mechanical properties.

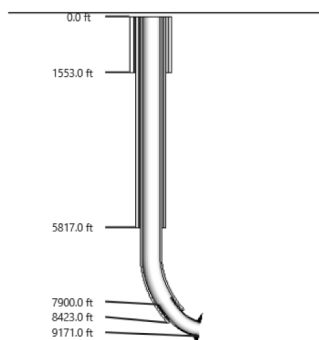


Figure 1—Well Schematic

12.25-in. Openhole Cement Operation Challenges

The 12.25-in. openhole section was planned with a 9 5/8-in. casing shoe to be set at 8,420 ft MD/8,245 ft TVD. Oil-based mud (OBM) was used to drill the section with a 40° inclination. Section challenges included loss circulation, narrow ECD margins for cementation, and the presence of CO₂, H₂S, and the risk of gas influx through the formation. OBM contamination can impact the cement strength; therefore, improved mud removal and effective water-wetting were required through spacers. Because of the ECD limitations, the slurry design required corrosion resistance at a lower density with advanced properties, full cement returns to the surface, and long-term zonal isolation were confirmed by CBLs. All objectives had to be met, and carbon emissions had to decrease.

8.5-in. Openhole Cement Operation Challenges

The 8.5-in. openhole section was planned with a 7-in. liner shoe to be set at 9,170 ft MD/8,618 ft TVD. Polymer mud was used to drill the section with an 87° inclination. Section challenges included loss

circulation, narrow ECD margins for cementation, and the presence of CO₂, H₂S, and the risk of gas influx through the formation. The cement system required CO₂ resistance, placement to the top of liner, and zonal isolation confirmed by CBLs.

Cement System Selection Overview

Low-Portland Cement System. As the oil and gas industry advances toward a decarbonized future, the design of resilient cement barriers with reduced-Portland cement content becomes essential. In line with this objective, a low-Portland cement system was selected for deployment. These cement slurries are engineered with a significantly lower proportion of Portland cement compared to traditional systems, which reduces carbon emissions and delivers superior barrier reliability. Because the manufacture of Portland cement contributes substantially to global CO₂ emissions, the minimization of its use in well construction remains a critical strategy to achieve net-zero targets.

Improved Tailoring Capabilities. Through advanced formulation techniques, the mechanical properties of these systems are optimized through the combination of the synergistic behavior of engineered materials with Portland cement. The service company's proprietary design approach allows these low-Portland cement systems to deliver high compressive strength and ductility at reduced slurry densities, which helps ensure reliable long-term zonal isolation.

Lower Density, Higher Compressive Strength. The advanced synergy between specialized materials provides greater flexibility in system design compared to conventional cement formulations. While traditional high-density cements are often associated with superior mechanical performance, laboratory evaluations indicate that low-Portland systems can achieve a compressive strength-to-Young's modulus ratio higher than that of standard high-density slurries.

Increased Ductility. Increased ductility in cement systems strengthens their ability to survive cyclic loads imposed during well operations and reduces the risk of stress-induced failures that could compromise barrier integrity. Low-Portland cement systems exhibit superior toughness and flexibility compared to conventional designs. These properties allow the systems to withstand downhole pressure and temperature fluctuations throughout the well's life cycle, which helps ensure long-term zonal isolation.

Reduced Permeability. These systems are specifically tailored to achieve a significant reduction in permeability compared to traditional cement designs. A low-permeability cement sheath provides improved resistance to corrosive fluid and gas migration, which maintains integrity during extended service durations and delivers a durable barrier.

Local Materials. The adaptable formulation methodology also allows for the integration of locally sourced, natural, and recycled materials. This design flexibility reduces dependency on Portland cement, mitigates supply chain challenges, and supports sustainability objectives (Halliburton 2022). In addition, these systems are compatible with current cement additives and require no specialized surface equipment to deliver both operational efficiency and environmental benefits.

Self-Healing Elastomer. The self-healing elastomer additive provides a proactive solution through expansion upon contact with OBM or synthetic-based mud. It seals under annular channels and re-establishes a competent barrier. This swell action allows the cement sheath to withstand pressure greater than 2,500 psi per 10 ft (3 m) of annular space. Its improved elasticity further allows the barrier to endure repeated pressure cycles typical of unconventional well completions, which helps ensure long-term seal integrity (Halliburton 2016).

Corrosion-Resistant Resin-Cement System. Cement operations for CO₂ and H₂S applications introduce distinct challenges related to maintaining long-term barrier integrity and well integrity. Downhole conditions that fluctuate in temperature and pressure cycling, along with chemical interactions, pose significant risks to the durability of conventional cement systems. In particular, the formation of carbonic acid from CO₂ and water can progressively degrade standard Portland cement matrices. Therefore, cement systems deployed in CO₂ and H₂S environments should exhibit a strong resistance to CO₂-induced degradation (Halliburton 2020).

Resin-enhanced cement systems are specifically engineered to deliver superior chemical resistance against CO₂ exposure and improved mechanical performance. These systems are designed with low-Portland content to withstand cyclic load stresses, which preserves the long-term integrity of the cement barrier in difficult CO₂ applications.

Reduced Permeability. The resin-cement system provides exceptional corrosion resistance through its composite design, which integrates Portland-based cement with a specialized resin formulation. The cement component follows industry best practices, with Portland content deliberately minimized and supplemented with CO₂-resistant materials to achieve the desired performance standards. A precise blend of resin and cement significantly reduces system permeability and strengthens barrier integrity (Halliburton 2020).

The resin component improves system performance through a dual mechanism. First, a portion of the resin forms a protective film on the surface of the composite to create a coat that helps improve the bond and serves as a barrier against external chemical interactions. Second, resin microspheres infiltrate the cement matrix, fill pore spaces, and reduce effective porosity. This structure forms an internal adhesive layer that shields cement grains from CO₂-induced degradation. (Salib and Fakher 2024)

The combined effects of surface coating and internal matrix reinforcement result in a composite system with markedly reduced permeability. This system delivers superior corrosion resistance under both dry and aqueous supercritical CO₂ conditions, which helps ensure long-term durability of the cement barrier.

Improved Mechanical Properties Deliver Long-Term Barrier Integrity. The addition of resin into the cement formulation significantly improves elasticity and shear bond strength of the cement sheath compared with conventional systems. Stronger shear bonds create firm adhesion between the cement, casing, and formation, which reduces the likelihood of debonding and the risk of gas migration and well integrity loss. Furthermore, the resin-modified cement sheath exhibits increased resistance to micro-cracks, which allows it to better absorb and withstand mechanical stress encountered during cyclic injection operations. These improvements help ensure long-term durability and reliability of the barrier throughout the well's operational life.

Slurry Design and Test Overview

Low-Portland Slurry Design and Properties

Slurry tests were performed by the service company per API standards and verified in the operator's cement laboratory. Table 1 and 2 and Figures 1 through 4 show the results.

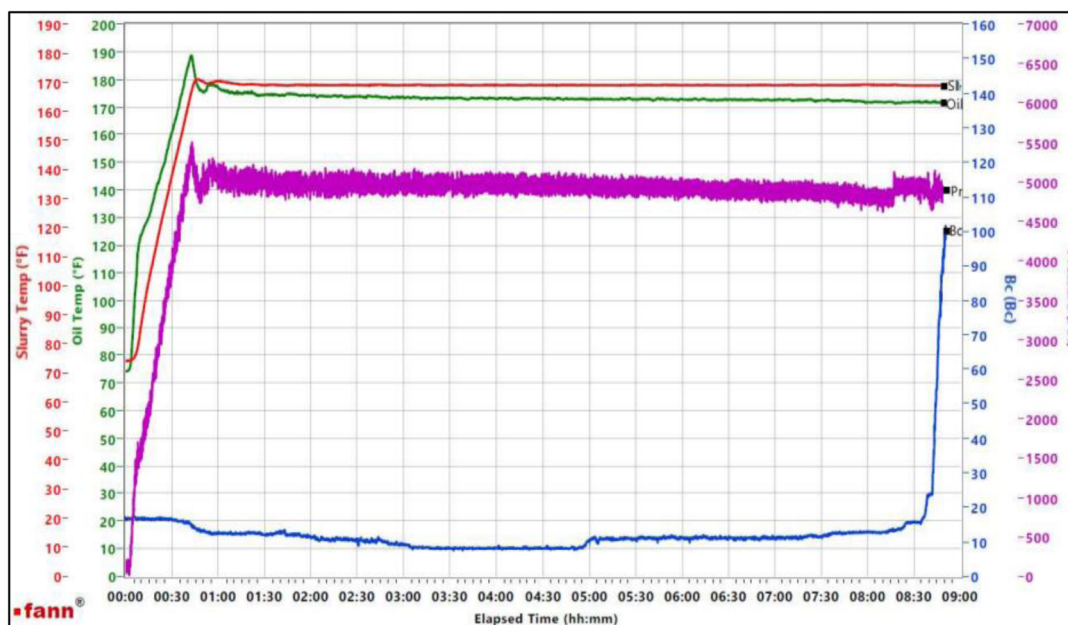
Table 1—Low-Portland slurry test results.

Slurry Design/Properties	Lead Cement	Tail Cement
Slurry Mixing Procedure	Fly-Mix	Batch-Mixed
Density (lbm/gal)	13.2	15.80
Free Fluid (ml)	0	0
API Fluid Loss (ml)	28	30
Thickening Time (hh:min)	08:50	06:47
UCA 24 hr (psi)	1533	2550

Table 2—Mechanical properties for 13.2 lbm/gal.slurry.

Sample Number	Test Date	Analysis Range [% of max. stress]	Compressive Strength [psi]	Young's Modulus [psi]	Poisson's Ratio [-]
1	6/8/2025	0.25 - 0.4	3800	1.13E+06	0.226
2	6/8/2025	0.25 - 0.4	3680	1.05E+06	0.207
3	6/8/2025	0.25 - 0.4	3719	1.01E+06	0.176
Average			3733	1.06E+06	0.203
Standard Deviation			61	5.80E+04	0.025
			1.6%	5.5%	12.4%

Sample Number	Test Date	Brazilian Tensile Strength [psi]
1	6/9/2025	419
2	6/9/2025	357
3	6/9/2025	392
Average		389
Standard Deviation		31
		8.0%

**Figure 2—Lead 13.2-lbm/gal thickening time chart**

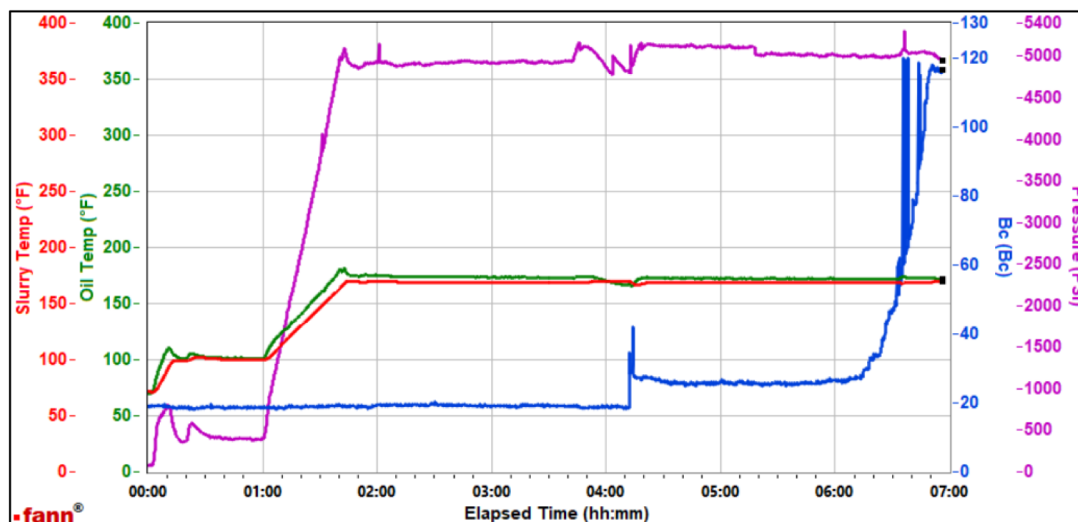


Figure 3—Tail 15.8-lbm/gal thickening time chart. (Test performed with 1 hr shutdown)

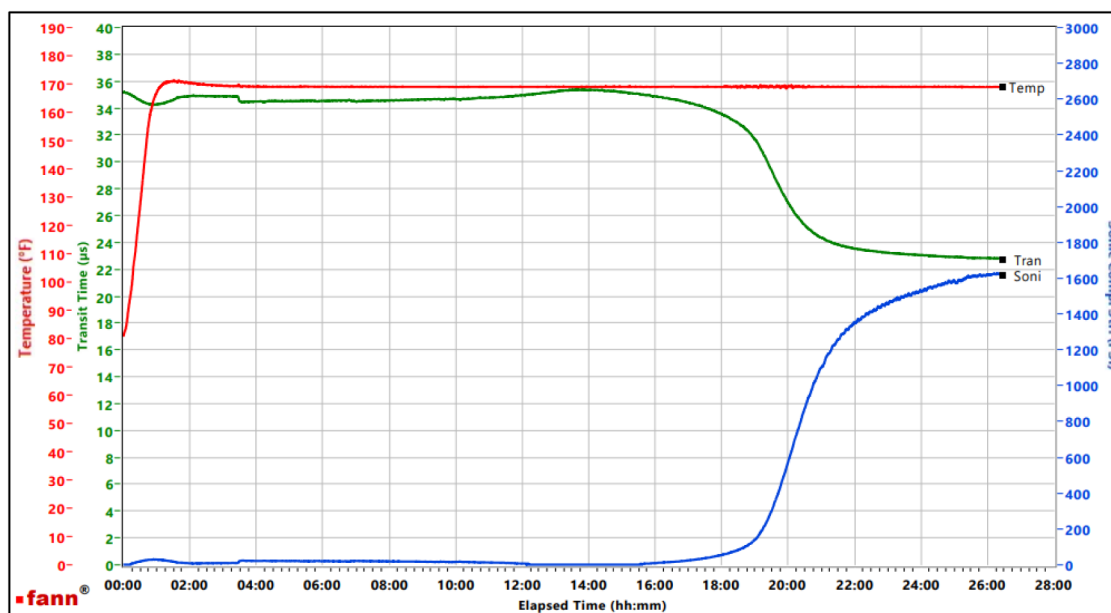


Figure 4—Lead 13.2-lbm/gal UCA chart.

Figure 2 is showing thickening time test done with 1 hour shutdown. Shutdown is included to simulate any unforeseen event that can occur during pumping of job.

Figures 4 and 5 illustrate the compressive strength expected after 24 hours. Figure 4 shows a reading of 1600 psi for the 13.2 ppg lead slurry, while Figure 5 shows a reading of almost 3000 psi for the 15.8 ppg tail slurry. This is a crucial design element to ensure that the well can withstand different loads during its construction

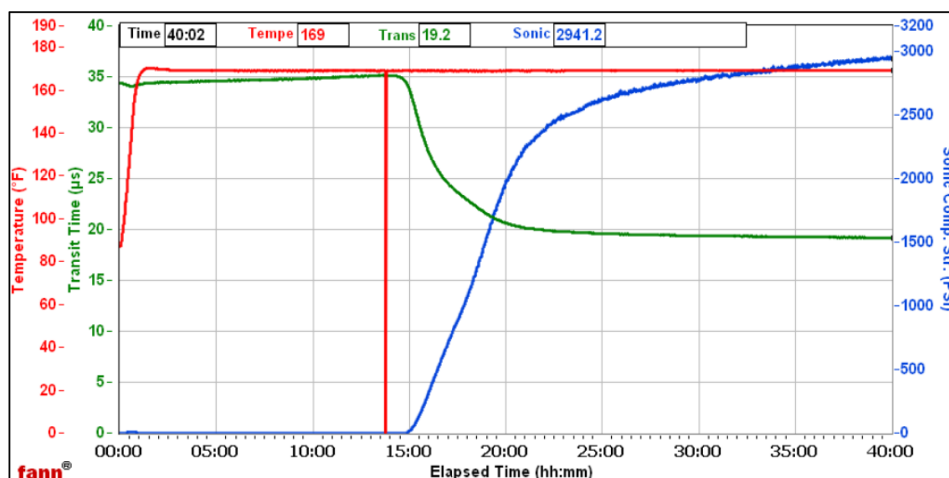


Figure 5—Tail 15.8-lbm/gal UCA chart.

CO₂-Resistant Resin-Cement Slurry Design and Properties

Slurry Design and Test Overview. Slurry tests were performed by the service company per API standards and verified in the operator's cement laboratory. Table 3 and Figures 6 and 7 show the results. Figure 6 is performed with 10% mud contamination and shows good compressive strength at end of test.

Table 3—CO₂-resistant resin-cement slurry testing results.

Slurry Design/Properties	Primary Cement
Slurry Mixing Procedure	Batch Mix
Density (lbm/gal)	15.80
Free Fluid (ml)	0
API Fluid Loss (ml)	28
Thickening Time (hh:min)	08:50
UCA 48 hr (psi)	3,246

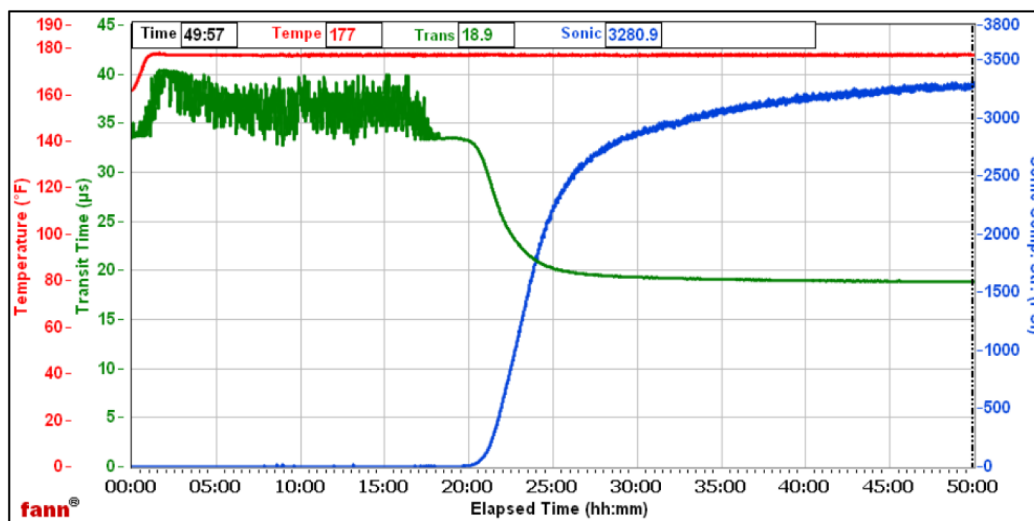


Figure 6—15.8-lbm/gal UCA chart.

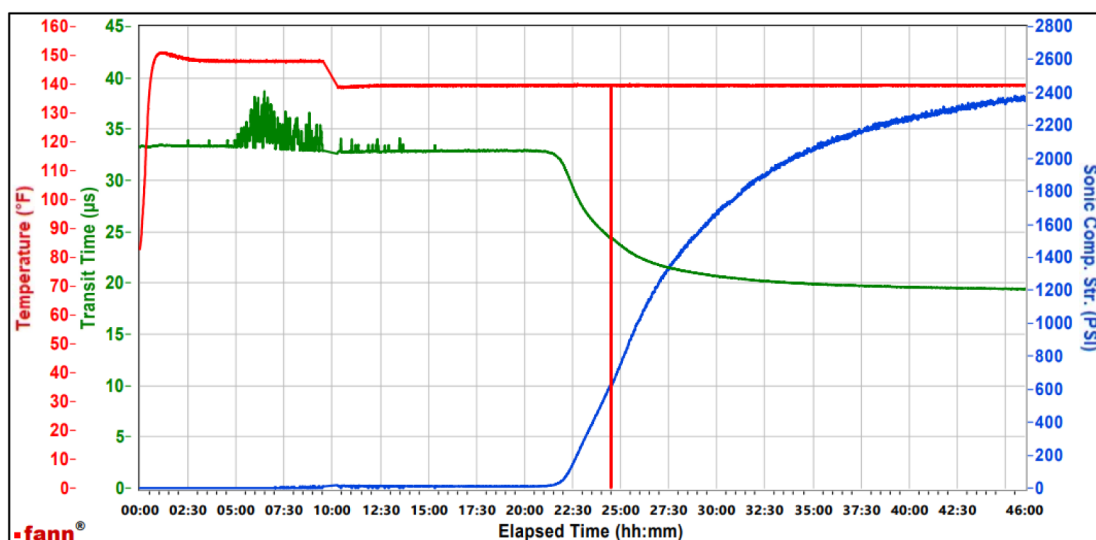


Figure 7—15.8-lbm/gal UCA chart with 10% mud contamination.

Mechanical Property Tests. Fig. 8 and Fig. 9 show the results of the mechanical property tests.

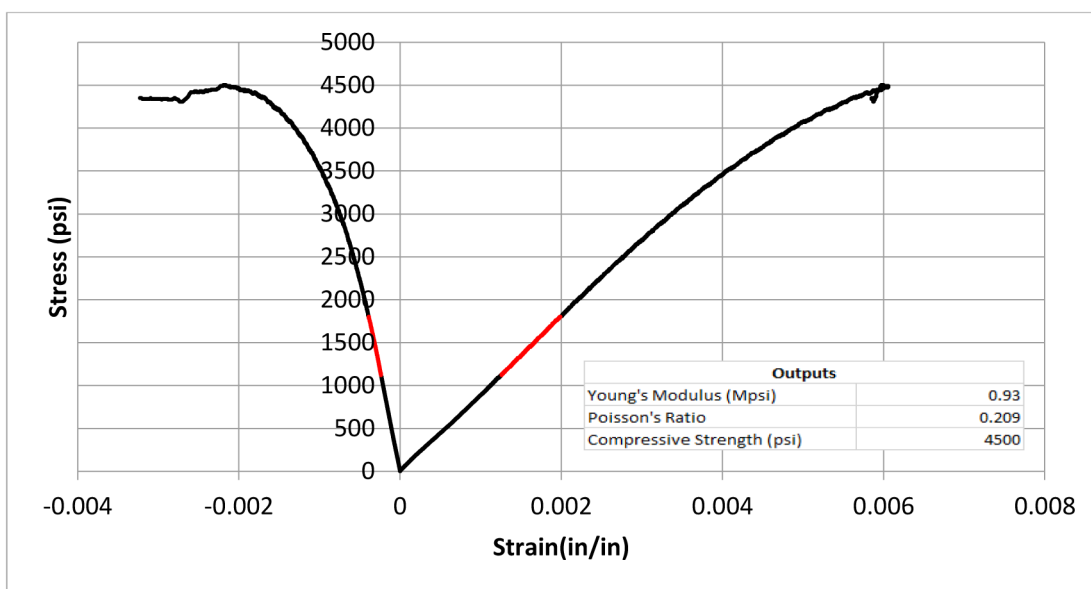


Figure 8—Mechanical property results (strain vs. stress).

Sample No.:		BR TS S-4, BR TS S-2, BR TS S-3, BR TS S-1		
Specimen Age:		7.00 day		
Temperature :		220 °F		
Pressure:		5,570 psi		
Method Name:		Brazilian Tensile Strength_2024		
Diameter in	Length in	Area in ²	Ultimate Force lbf	Brazilian Tensile S psi
2.00	1.08	3.29	676	199
2.00	1.02	3.20	600	187
2.00	0.980	3.08	594	193
2.00	0.980	3.08	567	184

Figure 9—Brazilian tensile strength.

The enhanced mechanical properties of this system, as shown in Figure 8, indicate a low Young's Modulus, which suggests greater flexibility compared to conventional rigid Portland cement. Additionally, Poisson's Ratio of 0.209 predicts the behavior of the cement system under multi-axial stresses, making it well-suited for the various stress and strain cycles experienced throughout the well's lifetime

Application

Operation Execution

From an operational standpoint, the cement operations for both the 9 5/8-in. casing and the 7-in. liner were executed without the requirement of additional equipment or specialized personnel. The current cement package on the rig site was fully used, with cement mixed in standard batch mixers and pumped through conventional cement pumps. A competent rig-site crew managed task distribution and helped ensure proper coordination of the cement mix, pump operations, and material supply throughout the operation.

Operation Log. The 9 5/8-in. casing cement operation followed this pump sequence: the operation commenced with circulation of 300 bbl of low- viscosity water-based mud at a high flow rate to condition and clean the wellbore. Next, 80 bbl of 11.7-lbm/gal of a lost circulation spacer was pumped, reinforced with fibers, at 5 bbl/min to help ensure effective mud removal and fluid interface stability. Subsequently, the bottom plug was dropped, and 410 bbl of 13.2-lbm/gal lead slurry were mixed and pumped on the fly at 5 bbl/min. This step was followed by 200 bbl of 15.8-lbm/gal batch-mixed tail slurry, also pumped at 5 bbl/min.

Displacement and 9 5/8-in. Casing Operation. Displacement began with the launch of the top plug, followed by 10 bbl of 15.8-lbm/gal tail slurry. The sequence continued with 20 bbl of spacer, after which the rest of the volume was displaced with 10.9-lbm/gal water-based mud at the maximum achievable rig pump rate. As operations approached the plug bump, 20 bbl short, the displacement rate was reduced to 3 bbl/min to help control pressure.

The top plug successfully bumped at 2,000 psi, and pressure held steady for 10 minutes to verify the integrity of the float equipment, which held pressure with no anomalies or leaks observed. A total of 74 bbl of uncontaminated lead cement returned to surface, which confirmed complete and proper annular cement coverage behind the casing.

Several critical success factors contributed to the effective execution of the 9 5/8-in. casing cement operation. The wellbore exhibited excellent integrity, with no losses observed before or during the operation, which helped ensured accurate volume control. Standoff was optimized to 85% through strategic placement of centralizers at every joint in the openhole section and every second joint in the cased interval, which improved circumferential cement sheath coverage.

The deployment of top and bottom plugs significantly minimized contamination risk because they acted as mechanical barriers and helped ensure proper fluid separation from other fluids in the wellbore. Moreover, a carefully designed density hierarchy in the mud, spacer, lead slurry, and tail slurry promoted efficient fluid displacement and reduced intermixing. A specially formulated spacer system further maximized mud removal efficiency and strengthened cement-to-formation bonds. Rig pumps delivered 98% efficiency, which allowed high-rate cement displacement and improved cement placement quality. Cement volumes were designed with 50% excess over openhole volume to help ensure returns at surface, which confirmed complete annular fill.

Upon completion, the cement shoe was drilled out with returns of hard cement observed. A subsequent shoe bond test (SBT) was performed at 0.65 psi/ft, which verified a strong and competent cement bond around the shoe.

7-in. Liner Cement Operation. The 7-in. liner cement operation faced partial losses, with dynamic losses recorded at 25 bbl/hr. Several loss circulation treatments failed, and the decision was made to proceed with the operation using all possible mitigations and operational precautions to help ensure effective zonal isolation. Cement volumes were designed with 70% excess over the openhole capacity to compensate for ongoing losses and to secure sufficient cement coverage.

The pump sequence began with a 50 bbl high-viscosity sweep pill to clean residual cuttings. The well was circulated through three full cycles to cool the formation and prepare the hole. Then 200 bbl of low-viscosity 10.4-lbm/gal water-based mud were pumped ahead of the cement operation to clean the hole further and remove any mud filter cake from the formation walls.

Next, 100 bbl of a specially designed fiber-laden spacer were pumped. The fibers acted as a bridge agent to reduce fluid loss, seal microfractures, and improve the bond between the cement and formation. Subsequently, 92 bbl of 15.8-lbm/gal corrosion-resistant resin-cement system slurry, reinforced with fibers, were batch mixed and pumped. A dart plug was launched, followed by 10 bbl of weighted spacer to push the cement ahead. Displacement continued using the cement unit to accurately monitor pressure and volume.

No definitive plug bump was observed during displacement. An additional 3 bbl (half of the liner shoe track volume) were pumped, but the plug still did not bump. Reverse circulation was initiated, and 20 bbl of uncontaminated cement were recovered at surface, which indicated sufficient cement placement.

The liner shoe track was later drilled out with medium-to-hard cement returns observed. An SBT was performed at 0.65 psi/ft, which successfully held and confirmed effective zonal isolation around the shoe.

Several key factors contributed to the success of the 7-in. liner cement operation. The wellbore was thoroughly cleaned through a combination of high-viscosity sweeps, low-viscosity mud circulation, and adequate preparation circulation cycles, which helped ensure optimal mud removal. Additionally, a fiber-laden spacer with sufficient contact time further improved wellbore conditioning.

Casing centralization was meticulously planned to achieve an 85% standoff through the installation of two centralizers per joint in the openhole section and one centralizer per three joints in the overlap (casedhole) section. Another critical enabler was the continuous rotation and reciprocation of the liner during cement displacement, which helped ensure full 360° cement sheath coverage, particularly in the horizontal sections.

Expected Emission Reduction

For a conventional well with a similar scope and drilling depth, estimated CO₂ emissions would typically reach approximately 1,024,796 kg of CO₂. In contrast, actual CO₂ emissions for this project measured 824,961 kg of CO₂, which represents a total reduction of 199,835 kg of CO₂. Several factors contributed to this reduction, such as optimized fuel consumption, cement system design, and drilling fluid selection. Cement operations alone accounted for approximately 8% of the total CO₂ reduction, equivalent to 49,819 kg of CO₂ saved.

To put this into perspective, the emission reduction achieved through cement operations alone is equivalent to the removal of approximately 43 cars from the road for one year. (Carbon footprint calculator n.d.)

Adoption of a low-Portland cement system in all well sections is strongly recommended to further improve CO₂ reduction and to support the achievement of long-term sustainability objectives.

Results and Verification

All cement quality parameters were successfully achieved for the 9 5/8-in. casing section. However, certain aspects of the 7-in. liner cement operation, such as loss management and productive plug bump, did not meet expectations. Nevertheless, the observation of pure cement returns in both operations confirmed effective displacement and suggested complete cement coverage behind the casing.

Cement bond evaluation on the 9 5/8-in. casing occurred 17 days after the cement operation. The log results confirmed excellent cement bond in the openhole interval from 7,900 to 5,100 ft (Fig. 10). However, the casing-to-casing section displayed some intervals of poor bond, likely a result of the double casing effect, a known factor that can impact log interpretation in such configurations.

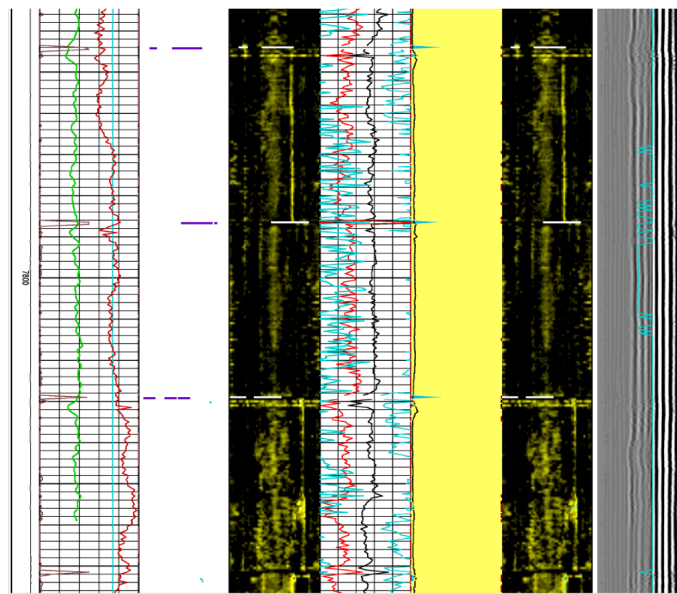


Figure 10—5/8-in. casing CBLs.

The cement bond evaluation for the 7-in. liner occurred 11 days after the cement operation. The results confirmed a continuous, high-quality bond in the entire evaluated interval, which verified effective zonal isolation (Fig. 11).

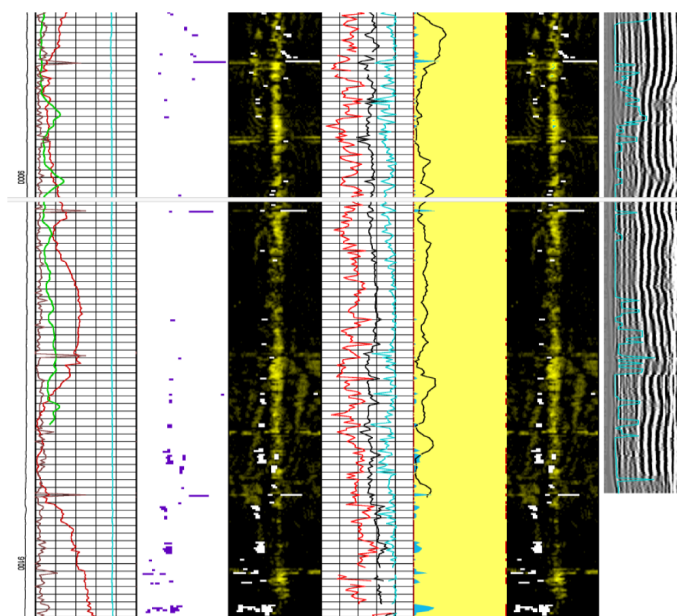


Figure 11—7-in. liner CBLs.

Conclusion

In the final analysis, the effective execution of low-Portland cement systems in this project represents a substantial advancement in carbon emission reduction while preservation of cement integrity remains

assured. The application of these reduced-Portland cement formulations in both the 9 5/8-in. casing and 7-in. liner yielded exemplary cement quality and helped ensure resilient zonal isolation. The incorporation of advanced cement methodologies, such as the deployment of specialized spacers and rigorous wellbore conditioning, further augmented system performance.

This project aligns strategically with the operator's 2030 Sustainability Strategy and the UAE's national decarbonization objectives and establishes a framework for future cement operations aimed at environmental sustainability and operational excellence. The measurable reduction in CO₂ emissions highlight the environmental benefits inherent in the adoption of low-Portland cement technologies. It is therefore recommended that these systems be progressively implemented in all well sections to achieve further carbon footprint reductions and contribute meaningfully to a sustainable and resilient energy sector.

Nomenclature

bbl/hr	barrels per hour
bbl/min	barrels per minute
CBL	cement bond log
CO ₂	carbon dioxide
ECD	equivalent circulating density
H ₂ S	hydrogen sulfide
lbm/gal	pounds per gallon
MD	measured depth
OBM	oil-based mud
TVD	total vertical depth

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